

Question 1 (No Code Required)

Scenario	Best Chart Type	Justification
1. Comparing total revenue across 6 product categories	Bar Chart (Amounts)	A bar chart is the best choice because it clearly compares values across different categories. It makes identifying the highest and lowest revenue easy.
2. Showing salary distribution across 4 job levels	Box Plot (Distribution)	A box plot shows median, spread, quartiles, and outliers, making it ideal for comparing salary distributions across job levels.
3. Showing market share of 5 competing brands	Pie Chart (Proportion)	A pie chart effectively shows how each brand contributes to the whole market share.
4. Relationship between advertising spend and sales revenue	Scatter Plot (x-y Relationship)	A scatter plot helps identify correlation, trends, and possible outliers between two continuous variables.
5. Monthly website traffic over 3 years	Line Chart (Temporal)	A line chart clearly displays trends and seasonal changes over time.
6. Employees reporting to managers	Network Diagram (Network)	A network (or organizational tree) diagram clearly represents reporting relationships between employees and managers.

Question 2

```
library(ggplot2)
```

```
# Dataset
```

```
df <- data.frame(
```

```
  Subject = c("Mathematics","Science","English","History","Computer Sci"),
```

```
  Average_Score = c(72,68,75,64,81)
```

```
)
```

```
# Sort descending
df <- df[order(-df$Average_Score),]

# Highlight highest score
df$Color <- ifelse(df$Average_Score == max(df$Average_Score),
  "Highest","Others")

# Preserve order
df$Subject <- factor(df$Subject, levels=df$Subject)

ggplot(df,
  aes(x=Subject,
    y=Average_Score,
    fill=Color))+

geom_col(width=0.7)+

geom_text(aes(label=Average_Score),
  vjust=-0.4,
  size=5)+

scale_fill_manual(values=c("Highest"="red",
  "Others"="steelblue"))+

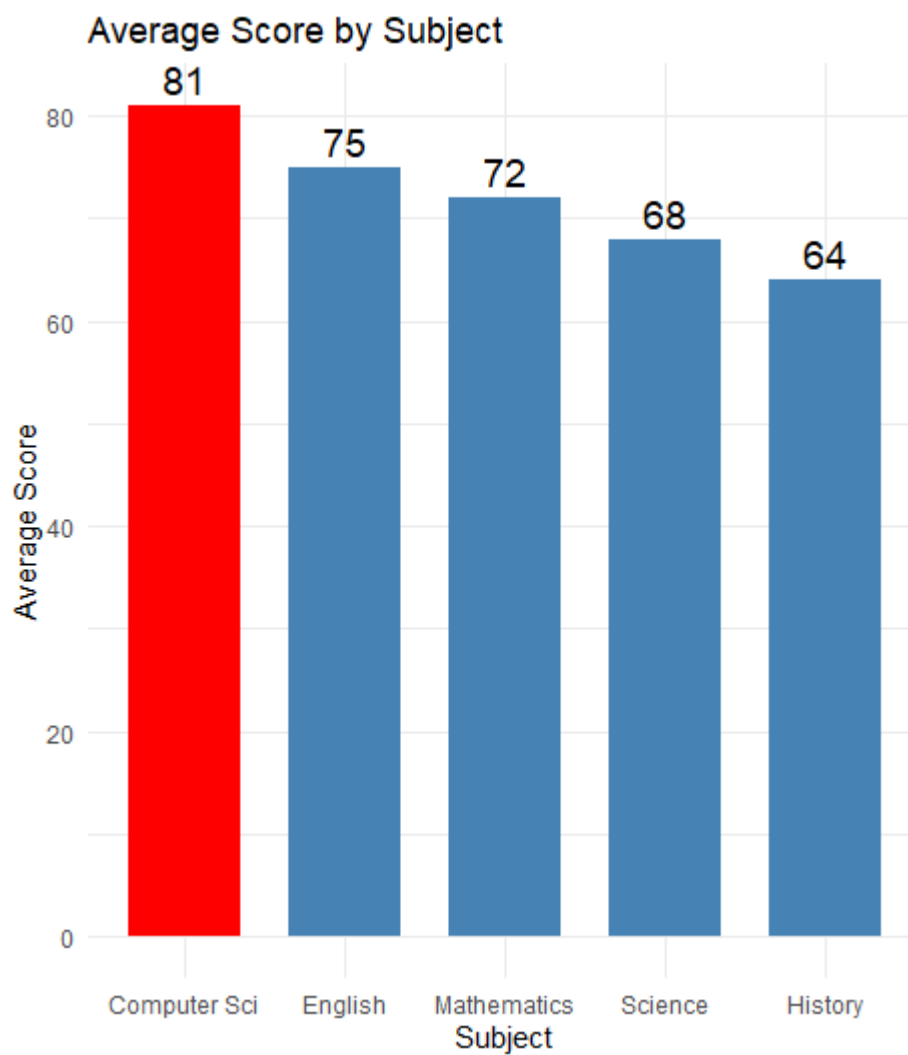
labs(title="Average Score by Subject",
  x="Subject",
  y="Average Score")+
```

```
theme_minimal()+
```

```
theme(legend.position="none")
```

Interpretation

- Computer Science has the highest average score (81).
- History has the lowest average score (64).
- The descending order makes comparison straightforward.



Question 3

```
library(ggplot2)
```

```
sales <- data.frame(
```

```
Region=c("North","North","North","North",  
         "South","South","South","South",  
         "East","East","East","East",  
         "West","West","West","West"),
```

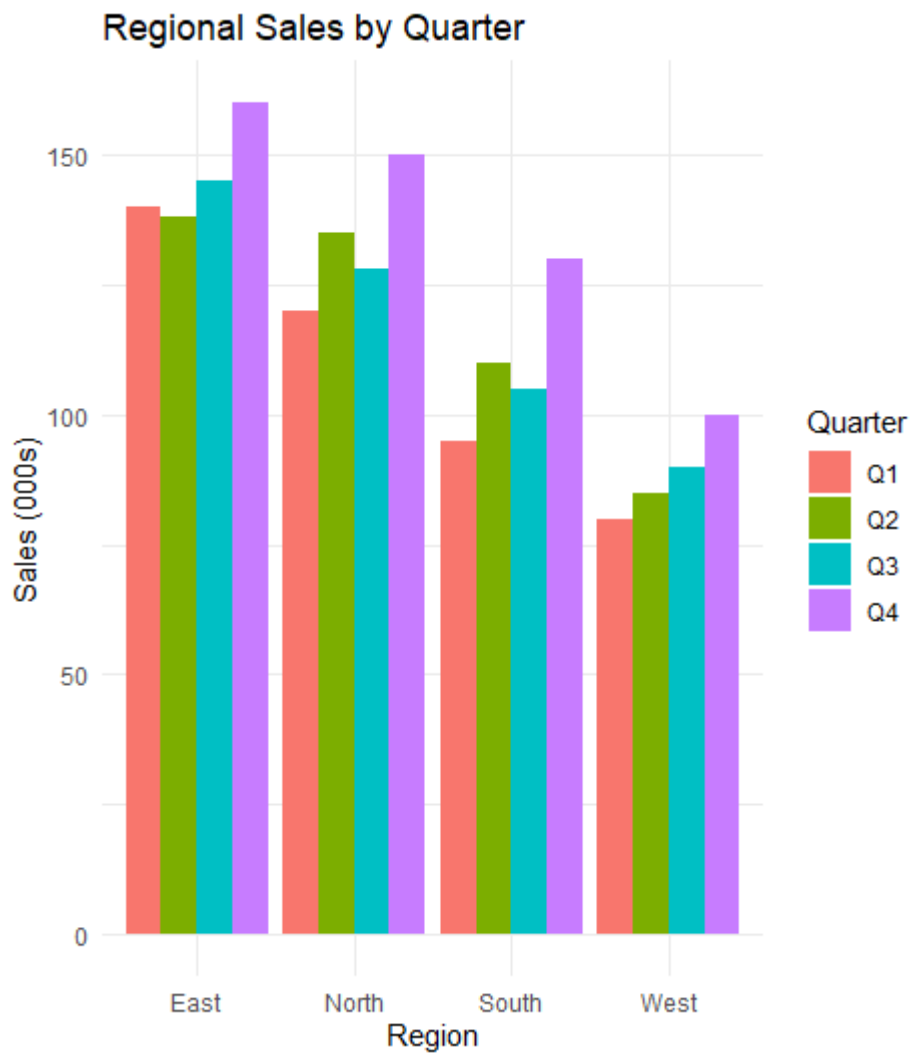
```
Quarter=c("Q1","Q2","Q3","Q4",  
          "Q1","Q2","Q3","Q4",  
          "Q1","Q2","Q3","Q4",  
          "Q1","Q2","Q3","Q4"),
```

```
Sales_000s=c(120,135,128,150,  
            95,110,105,130,  
            140,138,145,160,  
            80,85,90,100)  
)
```

(1) Grouped Bar Chart

```
ggplot(sales,  
       aes(x=Region,  
          y=Sales_000s,  
          fill=Quarter))+  
  
geom_col(position="dodge")+  
  
labs(title="Regional Sales by Quarter",  
     x="Region",  
     y="Sales (000s)")+
```

```
theme_minimal()
```



Interpretation

- East records the highest quarterly sales.
- West consistently has the lowest sales.

(2) Stacked Bar Chart

```
ggplot(sales,  
  aes(x=Region,  
    y=Sales_000s,  
    fill=Quarter))+
```

```
geom_col(position="stack")+
```

```
labs(title="Stacked Regional Sales",
```

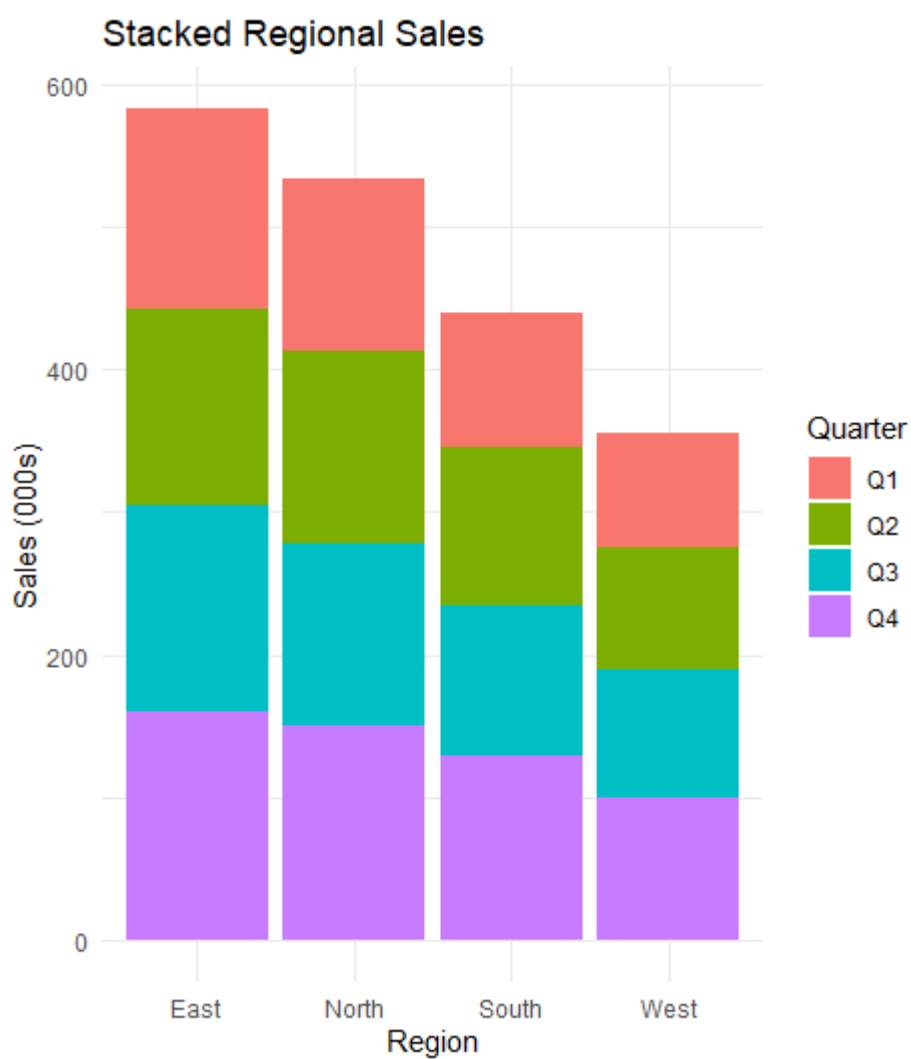
```
  x="Region",
```

```
  y="Sales (000s)")+
```

```
theme_minimal()
```

Interpretation

- East has the largest annual total.
- West contributes the least overall.



(3) Dot Plot

```
total <- aggregate(Sales_000s ~ Region,  
                   sales,  
                   sum)
```

```
total <- total[order(-total$Sales_000s),]
```

```
total$Region <- factor(total$Region,  
                       levels=total$Region)
```

```
ggplot(total,  
        aes(x=Region,  
            y=Sales_000s))+
```

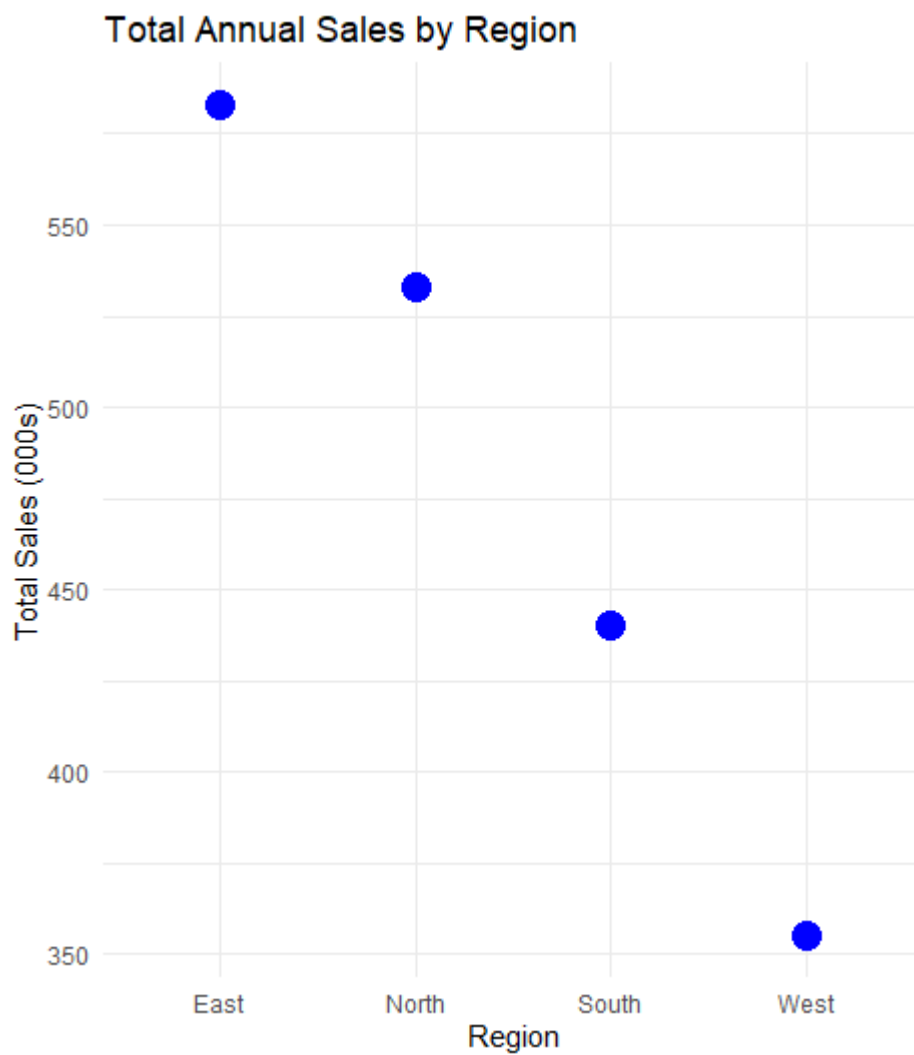
```
geom_point(size=5,color="blue")+
```

```
labs(title="Total Annual Sales by Region",  
     x="Region",  
     y="Total Sales (000s)")+
```

```
theme_minimal()
```

Interpretation

- East ranks first in annual sales.
- West ranks last.



(4) Heatmap

```
ggplot(sales,  
  aes(x=Quarter,  
    y=Region,  
    fill=Sales_000s))+  
  
  geom_tile(color="white")+  
  
  geom_text(aes(label=Sales_000s),  
    color="black")+  

```

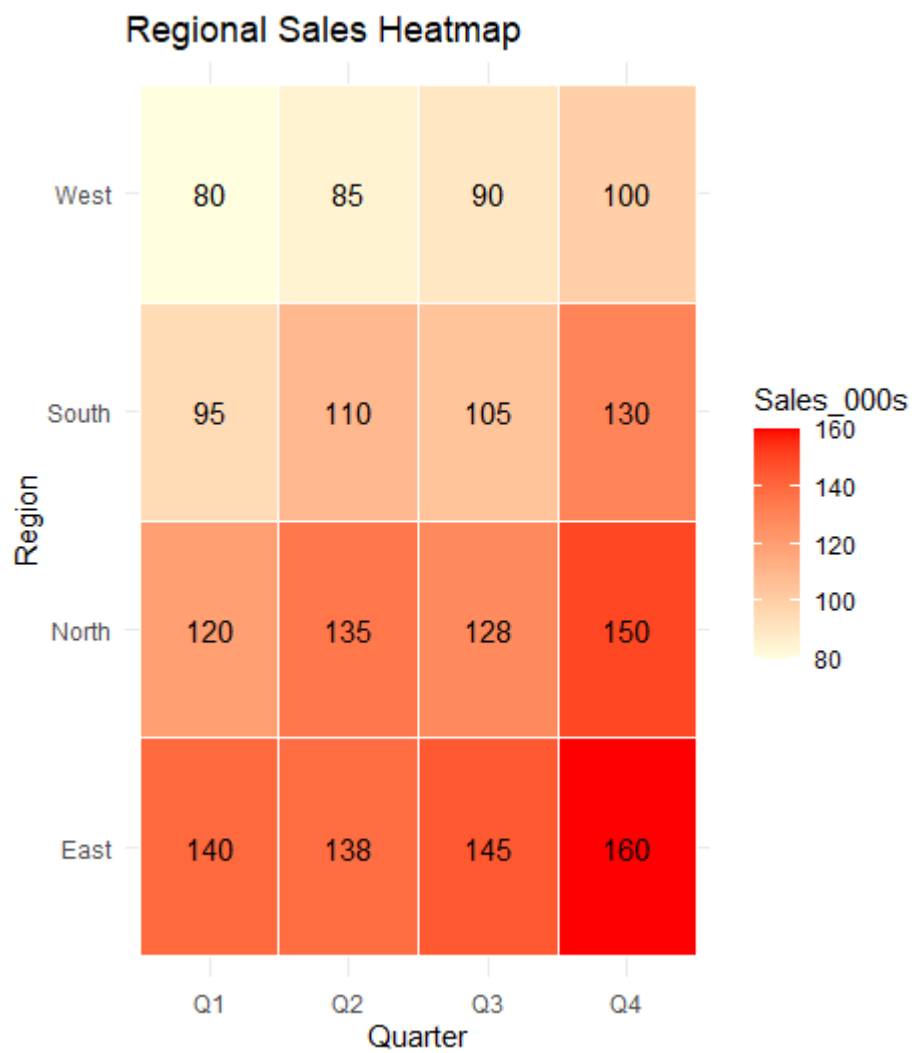
```
scale_fill_gradient(low="lightyellow",  
                    high="red")+
```

```
labs(title="Regional Sales Heatmap",  
     x="Quarter",  
     y="Region")+
```

```
theme_minimal()
```

Interpretation

- Darker colors indicate higher sales.
- East Q4 has the maximum sales (160).



Question 4

Generate Dataset

```
library(ggplot2)
```

```
library(ggribbles)
```

```
set.seed(42)
```

```
A <- rnorm(40,70,8)
```

```
B <- rnorm(40,75,5)
```

```
C <- rnorm(40,65,12)
D <- rnorm(40,80,6)
```

```
scores <- data.frame(
  Score=c(A,B,C,D),
  Class=rep(c("Class A",
    "Class B",
    "Class C",
    "Class D"),
    each=40)
)
```

(1) Violin Plot

```
ggplot(scores,
  aes(Class,
    Score,
    fill=Class))+
```

```
geom_violin(trim=FALSE)+
```

```
geom_boxplot(width=0.1,
  fill="white")+
```

```
labs(title="Exam Score Distribution by Class",
  x="Class",
  y="Score")+
```

```
theme_minimal()
```



Interpretation

- Class D has the highest median score.
- Class C has the greatest variability.
- Class B is the most consistent.

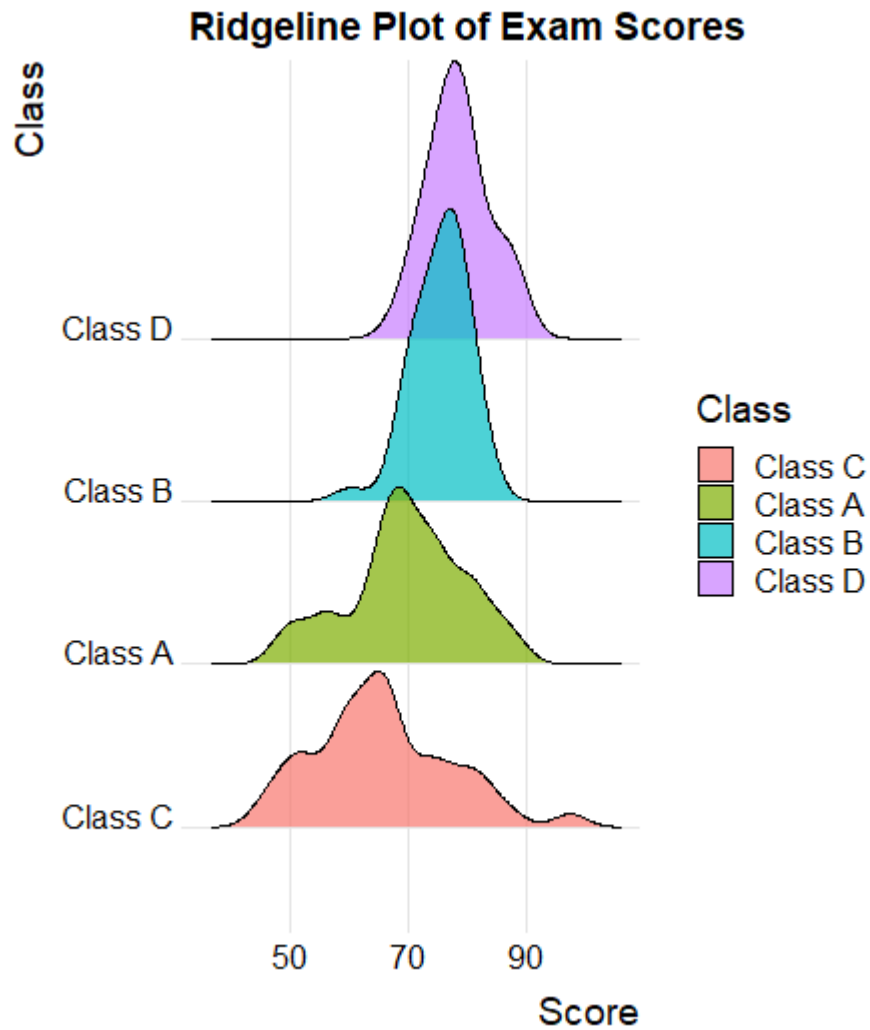
(2) Ridgeline Plot

```
means <- aggregate(Score ~ Class, scores, mean)
```

```
means <- means[order(means$Score),]
```

```
scores$Class <- factor(scores$Class,  
  levels=means$Class)
```

```
ggplot(scores,  
  aes(x=Score,  
    y=Class,  
    fill=Class))+  
  
geom_density_ridges(alpha=0.7)+  
  
labs(title="Ridgeline Plot of Exam Scores",  
  x="Score",  
  y="Class")+  
  
theme_ridges()
```



Interpretation

- The density for Class D is shifted furthest to the right, indicating the highest average score.
- Class C has the widest spread, showing greater variability.

Question 5

```
library(ggplot2)
```

```
library(reshape2)
```

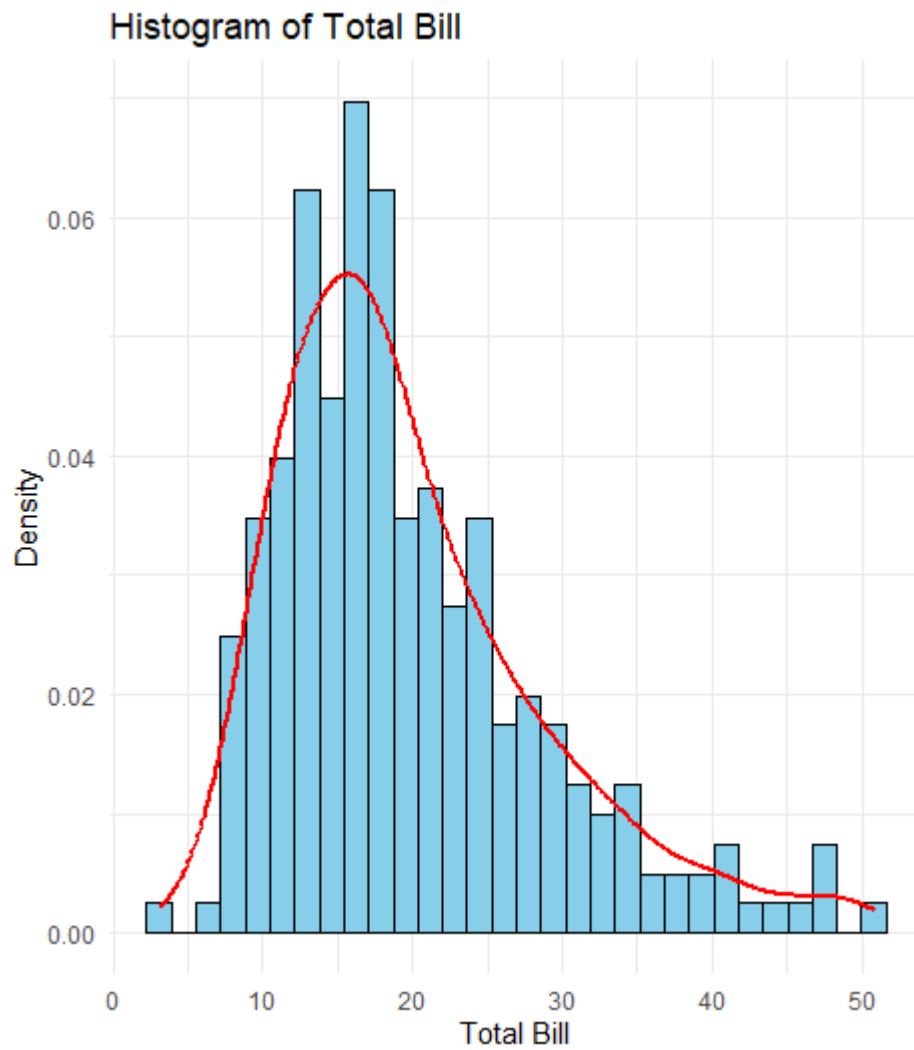
```
data(tips)
```

(1) Histogram with Density

```
ggplot(tips,  
  aes(x=total_bill))+  
  
  geom_histogram(aes(y=after_stat(density)),  
    bins=30,  
    fill="skyblue",  
    color="black")+  
  
  geom_density(color="red",  
    linewidth=1)+  
  
  labs(title="Histogram of Total Bill",  
    x="Total Bill",  
    y="Density")+  
  
  theme_minimal()
```

Interpretation

- Most bills fall between \$10 and \$25.
- The distribution is positively skewed with a long right tail.



(2) Histogram + Density by Smoker

```
ggplot(tips,  
  aes(x=total_bill,  
    fill=smoker))+  
  
geom_histogram(aes(y=after_stat(density)),  
  bins=30,  
  alpha=0.5,  
  position="identity")+
```

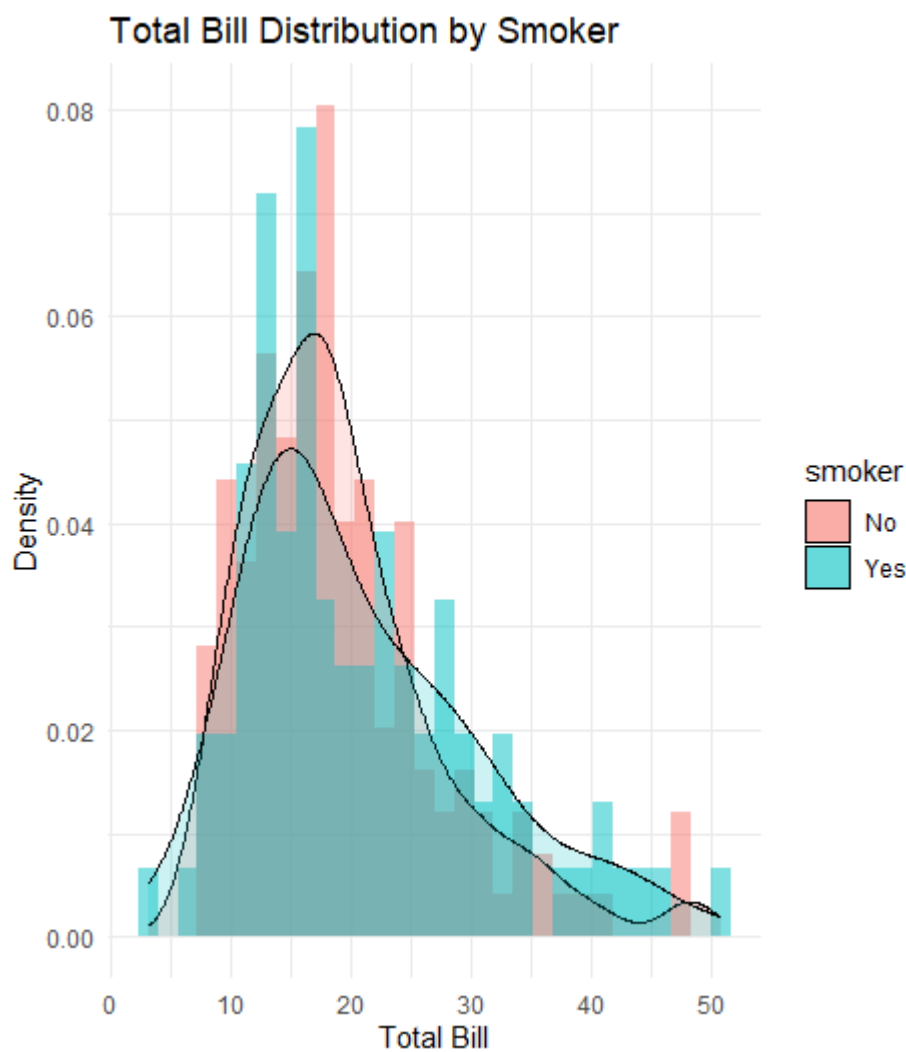
```
geom_density(alpha=0.2)+
```

```
labs(title="Total Bill Distribution by Smoker",
```

```
  x="Total Bill",
```

```
  y="Density")+
```

```
theme_minimal()
```



Interpretation

- Smokers generally show a slightly wider spread in total bills.
- Both smoker and non-smoker groups exhibit right-skewed distributions.
- The overlap indicates that the spending patterns are broadly similar, though smokers have a few higher-value bills.